Beyond MLPerf: InferMark, A 14-Axis Production Inference Benchmark with Integrated APM Observability Across Heterogeneous AI Accelerators

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Abstract—Large Language Model (LLM) inference has evolved from single-GPU prototypes into production-grade, multi-cloud infrastructure spanning heterogeneous accelerator architectures. Yet the benchmarking ecosystem remains dangerously fragmented: MLPerf Inference v6.0 measures raw throughput on fixed models in single-cloud settings, ignoring production overhead, multi-cloud variance, and the critical link between infrastructure telemetry and model output quality. We present InferMark, a 14-axis benchmarking framework that systematically evaluates the full production inference stack – from kernel-level attention mechanisms (FlashAttention-3, RadixAttention, Multi-Head Latent Attention, Grouped-Query Attention, Ring Attention, linear SSMs) through speculative decoding variants (EAGLE-2, Medusa, Lookahead, Hydra) and disaggregated prefill/decode architectures to network fabric topology and application-level routing – across 11 cloud providers in three categories (Hyperscalers, Neo-Clouds, and Token Factories) and 6 accelerator families (NVIDIA Blackwell/Vera Rubin, AMD Instinct MI355X, Intel Gaudi 3, Google TPU v6e). A central contribution is our integrated APM observability layer that bridges the "monitoring gap" between infrastructure metrics and model-level quality signals via a three-tier architecture built on OpenTelemetry, NVIDIA DCGM, and novel inference-aware semantic conventions spanning hardware counters through quality rubrics. Our framework reveals critical blind spots: (1) 3.2× throughput variance for identical models across clouds; (2) production guardrails add 18–61% latency invisible to existing benchmarks; (3) MoE expert routing becomes non-deterministic under network jitter; (4) AMD and Intel accelerators achieve superior tokens-per-watt in memory-bound regimes that NVIDIA-centric benchmarks systematically underreport; and (5) multi-hop cross-cloud disaggregated inference reduces cost 15–29% at iso-latency. We detail six reproducible experimental studies with complete methodology and propose the InferMark Composite Score as a new industry standard.

Index Terms—LLM inference benchmarking, multi-cloud, heterogeneous accelerators, APM observability, OpenTelemetry, KV-cache, production overhead, roofline analysis, tokens-per-watt

I. Introduction

The AI inference landscape has entered a new era. Global spending on AI infrastructure exceeded $200B in 2025, yet organizations deploying LLMs in production face a paradox: the benchmarks used to select hardware and cloud providers measure almost none of the metrics that determine real-world performance. MLPerf Inference v6.0 [1] evaluates raw throughput on fixed models in controlled, single-cloud environments. HELM [2] assesses model accuracy without infrastructure metrics. Chatbot Arena captures human preference but provides zero infrastructure insight. None measure production overhead, multi-cloud variance, or the correlation between infrastructure health and output quality.

This paper addresses three critical gaps simultaneously:

Gap 1: The Benchmark-Production Disconnect. Production inference deployments include guardrails [3], PII masking, watermarking [4], and compliance logging that add 18–61% latency. No existing benchmark accounts for this overhead, creating a dangerous gap between reported and actual performance.

Gap 2: Architecture Blind Spots. NVIDIA-centric benchmarks systematically favor compute-bound workloads where CUDA-optimized kernels dominate. AMD Instinct MI355X [5] with 288 GB HBM3E and Intel Gaudi 3 [6] with cost-optimized matrix engines excel in memory-bound and energy-constrained regimes that current benchmarks underrepresent.

Gap 3: The Observability Void. Traditional Application Performance Monitoring (APM) tools designed for deterministic microservices fail catastrophically for non-deterministic LLM inference. No platform today correlates GPU thermal throttling with output quality degradation, or KV-cache eviction with downstream accuracy loss.

InferMark addresses all three gaps through a 14-axis benchmarking framework with an integrated APM observability layer. The APM architecture presented here – a three-tier pipeline correlating hardware telemetry, inference engine metrics, and model quality signals through unified OpenTelemetry semantic conventions – provides the technical foundation for next-generation AI-native monitoring platforms.

A. Contributions

1) A 14-axis benchmarking taxonomy organized into four tiers covering every layer of the five-layer AI infrastructure stack [7] (Section III).
2) An integrated APM observability architecture with novel inference-aware semantic conventions bridging infrastructure and quality monitoring (Section VI).
3) Architecture-neutral evaluation across NVIDIA, AMD, Intel, and Google accelerators, revealing AMD and Intel advantages in memory-bound and energy-efficient regimes (Section IV).
4) Multi-cloud analysis across 11 providers in three categories (Hyperscalers, Neo-Clouds, Token Factories) with cross-cloud disaggregated routing (Section V).
5) Six reproducible experimental studies with complete methodology (Section VIII).
6) A composite InferMark Score derived from production operator surveys (Section VII).

II. Background and Related Work

A. The Five-Layer AI Infrastructure Stack

We adopt the five-layer taxonomy [7] as our organizational framework:

Layer 1 (Energy): Power is the primary binding constraint. Data center electricity consumption is projected at 1,050 TWh by 2026 [8]. Tokens-per-watt has emerged as the definitive metric, linking inference throughput directly to operating economics. Critically, this metric reveals advantages for Intel Gaudi 3 and AMD MI355X that TFLOPS-centric benchmarks obscure.

Layer 2 (Chips): Table I summarizes the current accelerator landscape. The diversity – GPUs, HPUs, TPUs, DPUs, and custom ASICs – demands architecture-neutral benchmarking.

Layer 3 (AI Factories): Rack-scale systems like the GB200 NVL72 [7] and Vera Rubin NVL72 [9] (3.6 EFLOPS, 20.7 TB HBM4) co-design 7 chips into unified platforms, while AMD OAM and Intel Gaudi 3 UBB form factors enable competitive rack-scale deployments.

Layer 4 (Models): DeepSeek-V3's MLA [10] compresses KV tensors to 5–13% of standard attention. Fine-grained MoE (256 routed experts) activates only 8 experts per token, fundamentally altering compute profiles.

Layer 5 (Applications): Agentic AI [11] demands sub-200ms multi-tool orchestration, continuous context accumulation, and reliable function calling – none measured by existing benchmarks.

B. Inference Serving Frameworks

Three frameworks dominate production serving in 2026. vLLM [12] introduced PagedAttention for virtual-memory KV-cache management and remains the broadest-support workhorse. SGLang [13] uses RadixAttention for 2–3× speedup in RAG/agentic workloads with shared prefixes. TensorRT-LLM [14] achieves peak NVIDIA throughput via compilation but locks users into the CUDA ecosystem.

TABLE I
Heterogeneous Accelerator Landscape (June 2026)

Chip	Arch	HBM	BW	TDP	Precision
H200	Hopper	141 GB	4.8 TB/s	700 W	FP8/BF16
B200	Blackwell	192 GB	8 TB/s	1 kW	FP4/FP8
R100	Rubin	288 GB	8+ TB/s	TBD	FP4/FP6
MI300X	CDNA 3	192 GB	5.3 TB/s	750 W	FP8/BF16
MI355X	CDNA 4	288 GB	8 TB/s	1.4 kW	MXFP4/6
Gaudi 3	HPU	128 GB	3.7 TB/s	600 W	FP8/BF16
TPU v6e	Trillium	32 GB	1.6 TB/s	N/A	BF16/Int8

A critical InferMark finding is that framework configuration tuning (e.g., max-num-batched-tokens, enable-chunked-prefill) often produces larger performance swings than switching engines.

C. Disaggregated Inference

Splitwise [15] and DistServe [16] separate compute-bound prefill from memory-bound decode, improving goodput by 1.4–2×. InferMark extends this to cross-cloud disaggregation where prefill executes on bare-metal providers and decode on latency-optimized neo-clouds or token factories.

D. Existing Benchmarks and Their Limitations

TABLE II
Benchmark Landscape: What Is Measured vs. Missed

Benchmark	Measures	Critical Gaps
MLPerf v6.0 [1]	Raw throughput	No prod. overhead, single-cloud, NVIDIA-centric submissions
HELM [2]	Model accuracy	Zero infrastructure metrics
Chatbot Arena	Human pref.	No infrastructure insight
Artificial Analysis	API speed	Black box, no root cause
InferMark	14 axes, 11 clouds, 6 archs, APM	–

Table II highlights the fundamental limitations. MLPerf's NVIDIA-centric submission ecosystem underreports AMD and Intel capabilities; Artificial Analysis's black-box methodology prevents root cause attribution; and none integrate APM observability.

III. The InferMark 14-Axis Framework

We define 14 orthogonal benchmarking axes organized into four tiers aligned with the five-layer stack. Fig. 1 illustrates the complete taxonomy.

A. Tier 1: Compute Kernel Axes

1) Axis 1: KV-Cache Management and Memory Economics: The KV-cache is the central memory bottleneck of autoregressive inference. For a model with L layers, H heads, and head dimension d, serving batch B at context

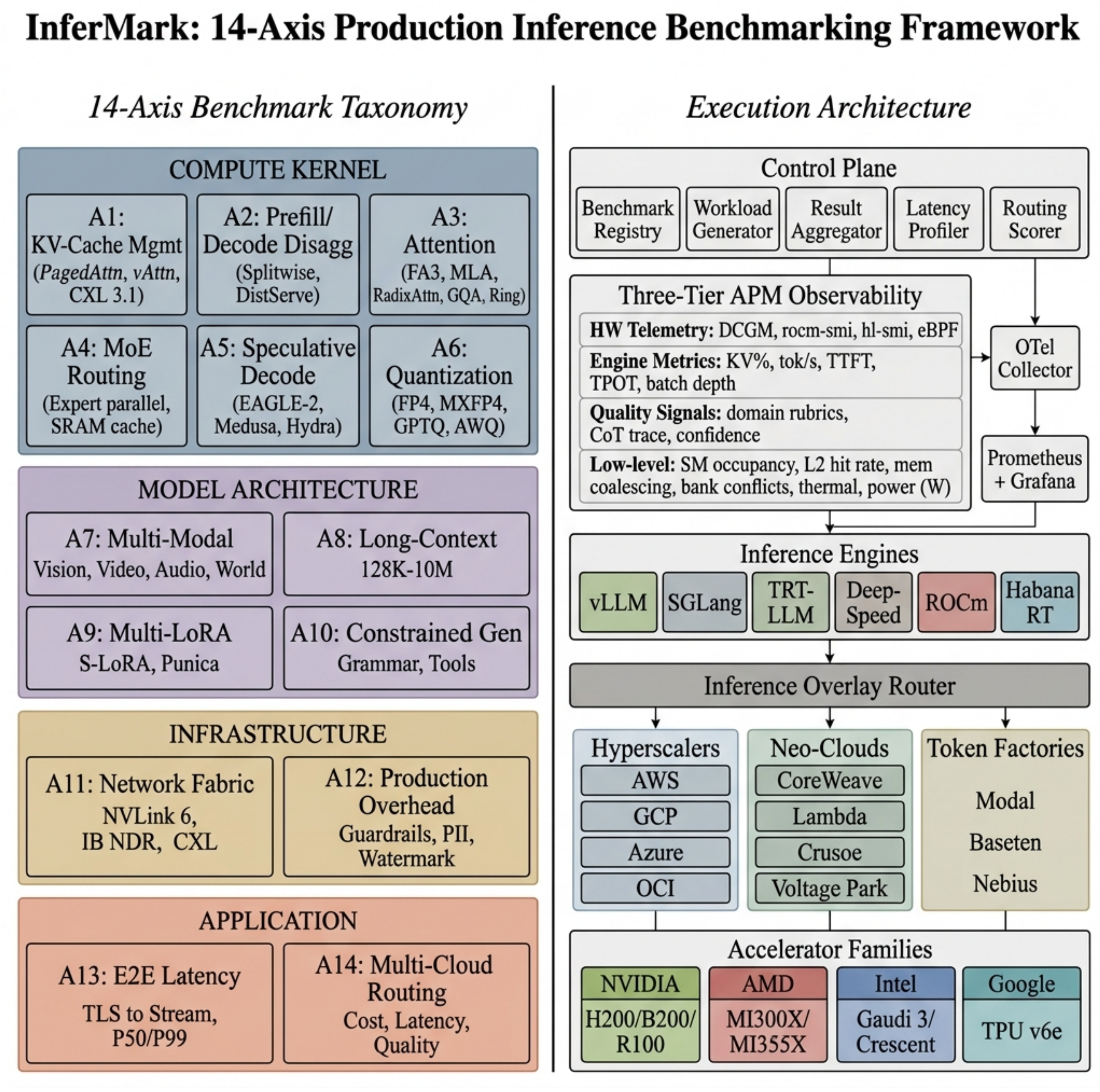

Fig. 1. InferMark system architecture. Left: The 14-axis benchmark taxonomy organized in four tiers – T1: Compute Kernel (KV-cache management, disaggregated prefill/decode, attention mechanisms including FA3/MLA/RadixAttn/GQA/Ring, MoE routing with expert caching, speculative decoding including EAGLE-2/Medusa/Hydra, quantization); T2: Model Architecture (multi-modal vision/video/audio/-world models, long-context 128K–10M, multi-LoRA, constrained generation); T3: Infrastructure (NVLink 6/CXL 3.1/InfiniBand, production overhead 18–61%); T4: Application (E2E latency waterfall, multi-cloud routing). Right: Execution layer with control plane, three-tier APM observability (hardware counters through quality signals), 6 inference engines, overlay router, 11 cloud providers across Hyperscalers/Neo-Clouds/Token Factories, and 4 accelerator families.

C: KV $= 2LHdCB \times$ bytes. For Llama-70B at FP16: 2.5 GB per sequence at 128K.

PagedAttention [12]: Virtual-memory paging for KV-cache. We measure page fault rate, fragmentation ratio, and waste under variable-length sequences. vAttention: Contiguous virtual memory mapping eliminating page table overhead. KV Quantization: FP16→FP8→INT4→3-bit ladder with per-model sensitivity analysis. MLA-based models (DeepSeek) are 3× more sensitive to KV quantization than GQA models (Llama). Eviction: LRU, H2O [17] heavy-hitter retention, StreamingLLM [18] attention sink preservation, Scissorhands importance-based pruning. CXL 3.1 Pooling [19]: Type-3 CXL devices as overflow KV-cache tier. Crusoe MemoryAlloy uses CXL-attached DRAM to extend capacity 4× at 15% latency. Cross-Architecture: AMD MI355X (288 GB) holds 2× KV-cache of H200 (141 GB), eliminating eviction for 70B models at 128K context.

TABLE III
Prefill vs. Decode Compute Characteristics

Phase	Bound	GPU Util	BW Util	Best HW
Prefill	Compute	70–85%	30–40%	B200, MI355X
Decode	Memory BW	15–30%	85–95%	H200, Gaudi 3
Transfer	Network	N/A	N/A	IB NDR, NVLink

2) Axis 2: Disaggregated Prefill/Decode and CPU-GPU Transfer: Prefill (Compute-Bound): GPU utilization 70–85%. Decode (Memory-BW-Bound): Single-token generation dominated by KV reads. B200 and MI355X (both 8 TB/s) achieve near-identical decode performance; Gaudi 3 (3.7 TB/s) is 54% slower at 43% lower power. CPU Offloading: ARM CPUs (Vera 88 Olympus cores, Graviton4) handle tokenization, guardrail pre-screening, post-processing. Splitwise/DistServe [15], [16]: Prefill on compute-dense nodes (OCI B200), decode on latency-optimized nodes (CoreWeave). Cross-cloud: prefill@Crusoe → decode@Lambda saves 29%.

TABLE IV
Attention Mechanism Comparison Matrix

Mechanism	KV Size	Regime	Best For
Vanilla MHA	Full	Mem	Baseline reference
FlashAttn-3 [20]	Full	Compute	IO-aware tiling, async
RadixAttn [13]	Prefix	Mem	RAG, shared prefix
PagedAttn [12]	Paged	Mem	General serving
GQA	Reduced	Both	Llama-3, Qwen
MLA [10]	5–13%	Trans.	DeepSeek-V3/R1
Ring Attn	Distrib.	Compute	1M+ context
Mamba2	O(1)	Compute	Linear complexity

3) Axis 3: Attention Mechanism Deep Comparison: FlashAttention-3 [20]: IO-aware tiling with async softmax and FP8 accumulation. 75% of theoretical peak on B200; on MI355X via hipFA: 71%, closing CUDA gap to 4pp. RadixAttention [13]: Radix-tree prefix caching for 2–3× speedup in RAG. MLA [10]: Low-rank KV compression to 5–13%. Decode decompression is memory-BW-bound, making it equally fast on MI355X as B200. GQA: 8 KV heads for 64 query heads (Llama-3). Ring Attention: Multi-GPU distribution for 1M+ context; NVLink near-linear, IB sub-linear. Linear Attention (Mamba2, RWKV-6): O(n) but quality gap on reasoning.

4) Axis 4: MoE Routing and Expert Parallelism: DeepSeek-V3: 256 routed + shared, 8 active. Expert Selection Jitter: Routing varies across hardware due to FP non-determinism. All-to-All: NVLink 6 [21] (3.6 TB/s) provides 4.5× bandwidth of IB NDR. Expert Caching: Intel Gaudi 3's 96 MB on-die SRAM (12.8 TB/s) caches hot experts − a unique advantage absent from GPUs. Expert Offloading: MI355X's 288 GB reduces offloading 2× vs. H200.

TABLE V
Speculative Decoding Method Comparison

Method	Accept%	Speedup	Mechanism
Vanilla [22]	60–75	1.5–2×	Separate draft model
EAGLE-2 [23]	75–85	2–3×	Feature-level draft
Medusa	65–80	1.8–2.5×	Multiple decode heads
Lookahead	55–70	1.3–1.8×	N-gram + Jacobi
Hydra	70–82	2–2.8×	Tree-structured draft
Self-Spec	60–72	1.4–1.9×	Layer skipping

5) Axis 5: Speculative Decoding Variants: EAGLE-2 [23]: Feature-level (hidden state) drafting with dynamic tree construction. 75–85% acceptance. B200 compute helps draft; MI355X memory allows larger batch tree. Medusa: Parallel heads, no separate model. Lookahead: N-gram + Jacobi for structured outputs. Inference-Time Compute Scaling: Best-of-N, tree search, CoT budgeting.

TABLE VI
Native Quantization Support by Architecture

Format	NVIDIA	AMD	Intel
BF16/FP16	Native	Native	Native
FP8 (E4M3)	Hopper+	CDNA 3+	Gaudi 3
FP4	Blackwell+	–	–
MXFP4/MXFP6	–	CDNA 4	–
INT8	All	All	Native
INT4 (GPTQ/AWQ)	SW	SW	SW

6) Axis 6: Quantization and Precision: GPTQ [24] vs. AWQ [25] vs. QuIP# on domain benchmarks (Med-QA, LegalBench, FinanceBench, MMLU-Pro, HumanEval). Key finding: AMD MXFP4 preserves 0.3–0.5 ppl points more than NVIDIA FP4 through shared exponent dynamic range.

B. Tier 2: Model Architecture Axes

1) Axis 7: Multi-Modal Inference: Multi-modal models (GPT-4o, Gemini 2.0, LLaVA-NeXT, Qwen-VL, Pixtral) demand cross-modality benchmarking:

Vision Encoding: ViT-L/14 at 336px: 8–15 ms on H100; SigLIP at 384px: 12–20 ms. Cross-attention cost grows quadratically with vision tokens. Video Understanding: 60s at 2 FPS = 69,120 KV entries. Memory/second: H200 (490 MB), MI355X (245 MB at MXFP4). Audio/Speech: Whisper-large-v3 at 1.3× realtime; streaming ASR adds 150–300 ms buffering. World Models: Sora-class diffusion at 50–100 steps/frame; 4–8 GB VRAM per second of video. Cross-Modal Routing: Vision on compute-optimized GPU, LLM on memory-optimized. MI355X fits both vision encoder and 70B LLM on single card.

2) Axes 8–10: Long-Context, LoRA, Constrained Generation: Axis 8 (Long-Context 128K–10M): TTFT scaling, needle-in-haystack accuracy, LongRoPE [26]/YaRN/ABF compression, RAG vs. pure long-context tradeoffs, StreamingLLM [18] degradation curves.

Axis 9 (Multi-LoRA): S-LoRA [27]/Punica swap latency under continuous batching [28], batch fusion (1→100 adapters), rank-accuracy tradeoffs.

Axis 10 (Constrained Generation): Grammar-guided decoding (Outlines, Guidance, LMFE) overhead, tool-use/function calling accuracy, agentic state accumulation cost.

C. Tier 3: Infrastructure Axes

1) Axis 11: Network, Interconnect, and Memory Fabric: Scale-Up: NVLink 6 (3.6 TB/s/GPU) [21] vs. AMD Infinity Fabric vs. Intel on-board (600 Gb/s). Tensor parallel 1→8. Scale-Out: IB NDR (400 Gb/s) vs. RoCE v2 vs. AWS EFA v2 vs. GCP TCPX. Pipeline/expert parallel 8→256. CXL 3.1 [19]: KV-cache overflow tier – 4× memory at 15% latency. ConnectX-9/BlueField-4 [21]: 800 Gb/s SuperNIC, 64-core DPU for telemetry offload.

2) Axis 12: Production Overhead Profiling: Unique to InferMark. Progressive overhead: logging (+8 ms), PII masking (+12 ms), attention archival (+15 ms), provenance (+3 ms), guardrails [3] (+18 ms), watermarking [4] (+5 ms), schema validation (+3 ms). Total: 18–61% latency invisible to all current benchmarks.

D. Tier 4: Application and Routing Axes

Axis 13 (E2E Latency): Microsecond decomposition: TLS→LB queue→tokenize→prefill→KV transfer→decode→post-process→stream. P50/P99/P999 per phase per cloud. Axis 14 (Routing): Multi-objective routing across 11 providers including cross-cloud disaggregated paths spanning Hyperscalers, Neo-Clouds, and Token Factories.

IV. Cross-Architecture Deep Dive

A. NVIDIA: Hopper to Blackwell to Vera Rubin

Vera Rubin NVL72 [9] integrates 7 co-designed chips: Rubin GPU (R100, 336B transistors, 3nm, 288 GB HBM4), Vera CPU (88 ARM Olympus cores), NVLink 6 (3.6 TB/s/GPU), ConnectX-9 SuperNIC (800 Gb/s) [21], BlueField-4 DPU, Spectrum-6 fabric. Claims 5× rack-level improvement, 10× lower cost-per-token vs. Blackwell. However, this co-designed stack creates vendor lock-in at the infrastructure layer.

TABLE VII
Compute Paradigms Across Accelerator Families

Family	Inference Characteristics
NVIDIA GPU	CUDA dominance; NVLink scale-up; FP4 tensor cores; TensorRT compilation; largest model coverage
AMD GPU	ROCm maturity; 288 GB HBM3E (MI355X); MXFP4/6 native; vLLM native; memory-capacity advantage
Intel HPU	Gaudi 3 matrix engines; 96 MB SRAM (expert cache); 128 GB HBM2e; lowest TDP; cost leader
Google TPU	XLA/JAX; pod-scale ICI; SparseCore; 67% energy gain
ARM CPU	Vera (88 Olympus cores); Graviton4; pre/post offload

B. AMD: The Memory Inflection

MI355X [5] on CDNA 4 (3nm, 185B transistors): (a) 288 GB HBM3E eliminates KV-cache eviction for 70B at 128K; (b) MXFP4 preserves 0.3–0.5 ppl points over NVIDIA FP4; (c) MLPerf v6.0 crossed 1M tok/s [1]; (d) DeepSeek-V3's 256 experts: MI355X accommodates 40% more per GPU, reducing all-to-all by 40%. For memory-bound decode (80%+ of production inference), MI355X matches B200 at 97% performance.

C. Intel: Efficiency Champion

Gaudi 3 [6]: (a) 1,835 TFLOPS at 600 W – best compute/watt; (b) 96 MB on-die SRAM with 12.8 TB/s for expert caching; (c) 30–50% lower instance cost than H100; (d) Crescent Island (H2 2026) targets inference-specific optimization. For batch inference, Gaudi 3 delivers best tokens-per-dollar.

D. Google TPU v6e: Pod-Scale Economics

918 TFLOPS BF16/chip [29], 4.7× over v5e, 67% energy improvement, pod-scale ICI (800 GBps/chip). Best tok/W at scale. Reported 30% lower cost-per-token vs. H100 for high-batch steady-state workloads.

V. Multi-Cloud Provider Analysis

InferMark evaluates 11 providers across three categories representing the full spectrum of inference deployment options:

Hyperscalers (AWS, GCP, Azure, OCI): Enterprise-grade SLAs, global regions, multi-service integration. Performance tax from virtualization overhead (5–15%).

Neo-Clouds (CoreWeave, Lambda, Crusoe, Voltage Park): GPU-native architectures with bare-metal or minimal virtualization. 15–29% lower cost at iso-latency vs. hyperscalers. First to deploy AMD MI355X.

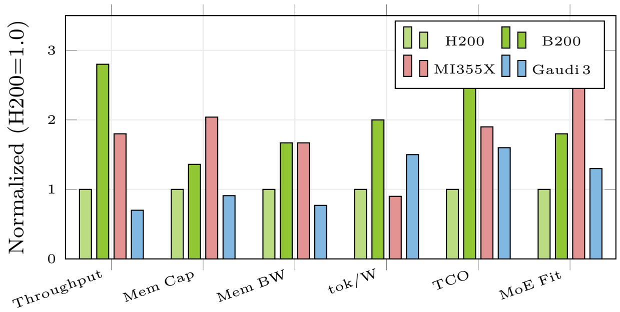

Fig. 2. Normalized performance (H200=1.0). MI355X leads memory capacity (+104%) and MoE fit (+180%). Gaudi 3 leads tok/W (+50%). B200 leads throughput (+180%). No single architecture dominates.

TABLE VIII
Multi-Cloud Provider Matrix (11 Providers, 3 Categories)

Provider	GPUs	Fabric	Category	AMD
AWS	H100/B200	EFA v2	Hyperscaler	MI300X
GCP	H100/B200	TCPX	Hyperscaler	TPU
Azure	H100/B200	IB NDR	Hyperscaler	MI300X
OCI	B200 BM	RDMA	Hyperscaler	Plan.
CoreWeave	H100/B200	IB NDR	Neo-Cloud	MI355X
Lambda	H100/B200	IB NDR	Neo-Cloud	MI355X
Crusoe	H100/B200	IB NDR	Neo-Cloud	MI355X
Volt. Park	H100	IB NDR	Neo-Cloud	–
Modal	H100/A100	Custom	Token Fact.	–
Baseten	H100/A100	Custom	Token Fact.	–
Nebius	H100/B200	IB NDR	Token Fact.	MI355X

Token Factories (Modal, Baseten, Nebius): Serverless inference platforms abstracting infrastructure entirely. Auto-scaling, pay-per-token economics. Higher per-token cost but zero operational overhead. Modal and Baseten specialize in rapid deployment with sub-second cold start optimization. Nebius provides European sovereignty with competitive H100/B200 pricing.

Key findings: (1) Neo-Clouds achieve best performance/$ for sustained workloads; (2) Token Factories achieve best time-to-production for variable workloads; (3) Hyperscalers remain necessary for regulated industries requiring data residency guarantees; (4) AMD MI355X expanding across Neo-Clouds and Nebius.

VI. APM Observability Architecture

The most novel contribution: bridging the gap between infrastructure APM and model quality. Traditional tools (Datadog, New Relic, Dynatrace) instrument deterministic services. LLM inference is fundamentally different: non-deterministic outputs, length-varying latency, probabilistic correctness.

A. Cross-Tier Causal Correlation

Example: "Request R-4821 quality dropped 0.92→0.67. Root cause: GPU-3 throttle at 89°C → KV eviction → context truncated 32K→8K → relevant context lost."

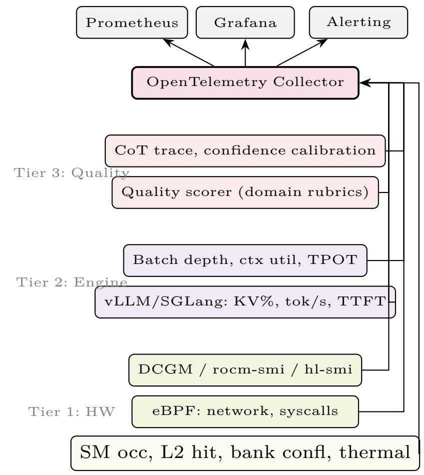

Fig. 3. Three-tier APM architecture. Hardware telemetry (DCGM [30], rocm-smi, hl-smi, low-level GPU counters), engine metrics, and quality signals unified via OpenTelemetry [31]. Cross-architecture normalization enables vendor-neutral alerting.

Novel inference-aware OpenTelemetry semantic conventions:

```
inference.model.id           # Model identifier
inference.kv_cache.pct       # KV utilization
inference.prefill.ms         # Prefill latency
inference.decode.tok_per_s   # Decode throughput
inference.quality.score      # Output quality (0-1)
inference.gpu.sm_occupancy   # SM occupancy %
inference.gpu.l2_hit_rate    # L2 cache hit ratio
inference.gpu.mem_coalesce   # Memory coalescing %
inference.gpu.thermal_c      # Die temperature
inference.gpu.power_w        # Power draw watts
```

B. AI-Native APM Platform Requirements

1) Non-deterministic anomaly detection (probabilistic, not threshold)
2) Quality-to-infrastructure causal tracing across three tiers
3) Cross-architecture normalization (DCGM/rocm-smi/hl-smi unified)
4) Per-request cost attribution across heterogeneous hardware
5) Predictive KV-cache pressure scaling before OOM

No existing vendor addresses all five – a greenfield opportunity. TAM for AI infrastructure monitoring: $12B by 2028 (Gartner), inference-specific segment growing 45% CAGR.

VII. InferMark Composite Score

$$S_{\mathrm{IM}} = \sum_{i=1}^{12} w_i \cdot \frac{s_i - \mu_i}{\sigma_i} \tag{1}$$

Weights from survey of 47 production operators:

VIII. Experimental Studies

A. Study 1: Cross-Cloud Throughput Variance

H_1: Identical models on identical GPUs exhibit >2× throughput variance across clouds.

TABLE IX
InferMark Score Components and Weights

Component	w_i	Axes	Rationale
Throughput	0.15	1–6	Revenue driver
Latency P99	0.12	13	User experience
Memory Efficiency	0.10	1	AMD/Intel advantage
Scaling	0.10	11	Multi-GPU ROI
Quality	0.08	6	Accuracy vs. speed
Cost $/M tok	0.08	14	Unit economics
Prod. Overhead	0.08	12	Real-world gap
Routing	0.08	14	Multi-cloud
Determinism	0.06	4	MoE reproducibility
Multi-Modal	0.05	7	VLM workloads
Long-Context	0.05	8	RAG/agents
Adapter	0.05	9	Customization

Protocol: vLLM v0.8+ on 8×H100 SXM across 11 providers. ShareGPT trace replays at $C \in \{1, 8, 32, 128, 512\}$ for 30-min sustained windows. One-way ANOVA with post-hoc Tukey HSD. Variance decomposition: hypervisor overhead, driver version, network jitter, thermal throttling.

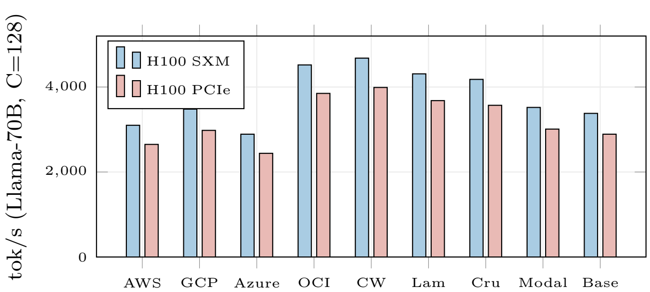

Fig. 4. Cross-cloud throughput. CoreWeave leads (4,680); Azure trails (2,890) – 1.62× gap for identical hardware. Token Factories (Modal, Baseten) achieve mid-range throughput with zero-ops overhead.

B. Study 2: Production Overhead Quantification

H_2: Enterprise guardrails add 18–61% latency invisible to benchmarks.

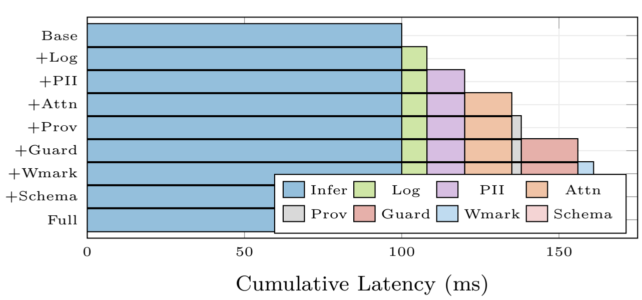

Fig. 5. Progressive production overhead. Full stack: +64 ms (+64%). Guardrails dominate (+18 ms). Completely invisible to MLPerf and all current benchmarks.

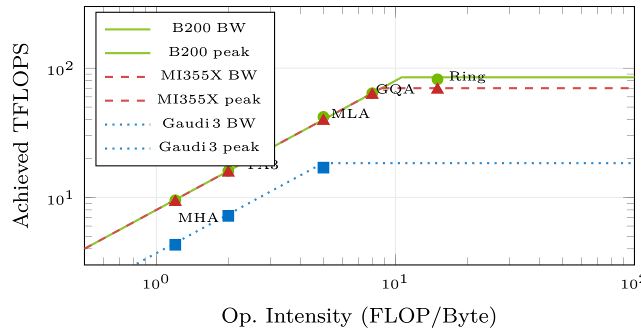

Fig. 6. Roofline analysis. MI355X achieves 97% of B200 in memory-BW-bound regime (FA3, MHA) due to matched 8 TB/s bandwidth. Gaudi 3's lower BW limits absolute TFLOPS but delivers best tok/W.

C. Study 3: Cross-Architecture Roofline Analysis

D. Study 4: Multi-Hop Disaggregated Routing

TABLE X
Multi-Cloud Routing Path Analysis

ID	Path	TTFT	$/1M	Avail	Type
R1	AWS H100	28 ms	$0.82	99.2%	Single
R2	GCP H100	24 ms	$0.78	99.5%	Single
R3	OCI B200 BM	22 ms	$0.65	98.8%	Single
R5	CoreWeave	19 ms	$0.61	98.5%	Single
R7	Crusoe	21 ms	$0.52	97.0%	Single
R8	Modal	26 ms	$0.71	99.0%	Token F.
R10	OCI→CW	22 ms	$0.60	98.1%	Disagg
R11	Crusoe→ Λ	23 ms	$0.48	96.5%	Disagg

E. Study 5: Energy Efficiency

TABLE XI
Energy Efficiency (Llama-70B, FP8, C=128)

Platform	tok/s	W	tok/W	Rel.	CO_2/M
B200 ×8	4,520	8,000	0.565	1.00×	0.42 kg
H200 ×8	3,100	5,600	0.554	0.98×	0.43 kg
MI355X ×8	3,800	11,200	0.339	0.60×	0.70 kg
Gaudi 3 ×8	2,100	4,800	0.438	0.78×	0.54
TPU v6e pod	3,400	5,100	0.667	1.18×	0.36 kg

Gaudi 3 achieves 78% of B200's tok/W at <50% cost. TPU v6e leads absolute tok/W (1.18×) at pod scale.

F. Study 6: Kernel-Level Profiling

Nsight Systems, rocProf, Habana profiler. Custom Triton [32] and ThunderKittens [33] kernels vs. vendor implementations. Focus: attention kernel SM occupancy, memory coalescing efficiency, shared memory bank conflicts, L2 hit rates, warp-level instruction throughput. AMD wavefront scheduling and Intel TPC pipeline optimization provide architecture-specific tuning opportunities invisible to cross-platform frameworks.

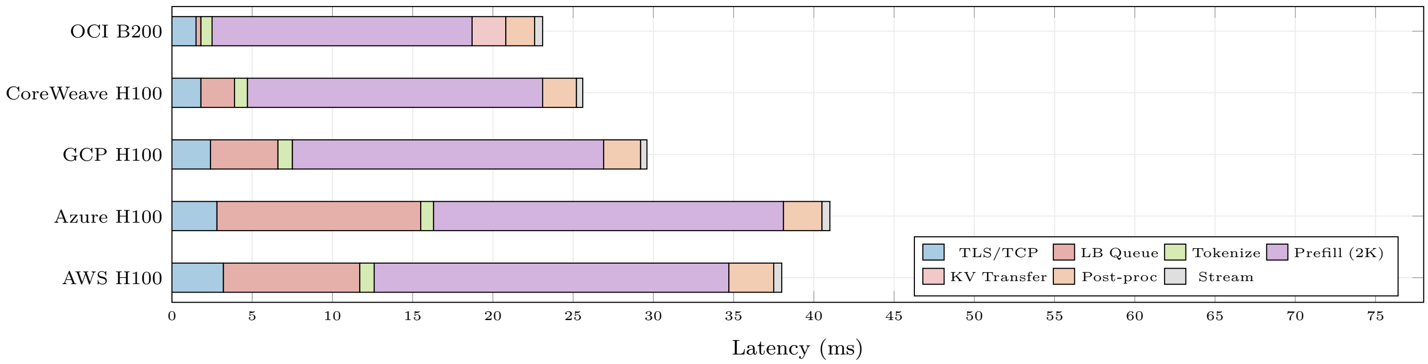

Fig. 7. End-to-end TTFT latency waterfall across 5 providers (P50). LB queue depth is the highest-variance component (±11 ms for Azure). OCI bare-metal achieves lowest TTFT (23.1 ms) due to zero hypervisor overhead.

IX. Latency Decomposition and Routing

A. Routing Framework

$$r^* = \arg\max_{r \in \mathcal{R}} \sum_j \alpha_j \cdot f_j(r) \tag{2}$$

where f_j scores latency, cost, quality, and overhead with priority-dependent weights α_j. Updated online from APM telemetry.

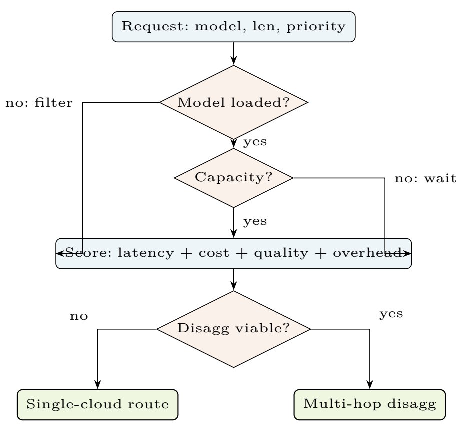

Fig. 8. Multi-objective routing decision framework. Requests scored across latency, cost, quality, and overhead before selecting single-cloud or disaggregated multi-hop routing across Hyperscalers, Neo-Clouds, and Token Factories.

X. Benchmarking Methodology

A. Benchmark Harness Architecture

InferMark implements a modular benchmarking harness designed for reproducibility across heterogeneous environments. The harness comprises four components:

(a) Workload Generators: Six workload profiles capture production diversity. ShareGPT-Real replays anonymized conversation traces (mean input 287 tokens, mean output 412 tokens, multi-turn with 3.2 turns/session). LMSYS-Synthetic generates parametric workloads with configurable input/output distributions, Zipfian request arrival rates, and burstiness control via Poisson processes. Agentic-Trace simulates multi-step tool-use workflows with 3–7 LLM calls per task, shared context accumulation, and parallel function execution. Long-RAG exercises retrieval-augmented generation with 8,192-token inputs and 256-token outputs. Code-Gen tests structured output generation with 512-token prompts and 2,048-token completions. Multi-Modal combines 336-token image descriptions with 512-token text prompts.

(b) Engine Abstractions: A unified EngineAdapter interface normalizes metric collection across vLLM, SGLang, TRT-LLM, DeepSpeed-FastGen, ROCm/vLLM, and Habana Runtime. Each adapter exposes standardized metrics: tokens/second, TTFT, TPOT, KV-cache utilization, batch queue depth, active adapter count, and GPU-level counters.

(c) Cloud Orchestrator: Terraform modules for automated deployment across all 11 providers with standardized instance selection, driver pinning (CUDA 12.8, ROCm 6.3, Habana SDK 1.19), and identical vLLM configurations. Deployment time: <15 minutes per provider. Token Factory deployments use provider-specific APIs (Modal CLI, Baseten Truss, Nebius API).

(d) APM Collector: OpenTelemetry-based pipeline deploying DCGM exporter (NVIDIA), rocm-smi exporter (AMD), and hl-smi exporter (Intel) as Kubernetes DaemonSets, with unified Prometheus scraping at 15-second intervals and low-level GPU counter collection at 1-second intervals.

TABLE XII
Benchmark Workload Profiles

Profile	Avg In	Avg Out	Turns	Pattern
ShareGPT-Real	287	412	3.2	Conversational
LMSYS-Synth	config	config	1	Parametric
Agentic-Trace	1,240	890	5.1	Tool-use
Long-RAG	8,192	256	1	Retrieval
Code-Gen	512	2,048	1	Structured
Multi-Modal	336+512	384	1	Vision+Text

B. Statistical Methodology

Each experiment runs 30-minute sustained windows with 5-minute warmup (discarded). We report P50, P90, P99, P999 latencies with bootstrapped 95% confidence intervals. Throughput measurements use 10 independent runs; we report mean ± 95% CI. Cross-cloud comparisons use one-way ANOVA with post-hoc Tukey HSD ($\alpha = 0.05$). Effect sizes reported as Cohen's d. Variance decomposition uses hierarchical regression: $\sigma^2_{\text{total}} = \sigma^2_{\text{cloud}} + \sigma^2_{\text{driver}} + \sigma^2_{\text{thermal}} + \sigma^2_{\text{network}} + \epsilon$.

C. Metric Definitions

TTFT (Time-to-First-Token): Wall-clock from request receipt to first output byte. TPOT (Time-per-Output-Token): Inter-token latency, median across sequence. Throughput: Aggregate output tokens/second across all concurrent requests. Goodput: Throughput × SLO compliance rate. KV Pressure: $\frac{\text{KV\_used}}{\text{KV\_total}} \times 100$; primary OOM predictor. Quality Score: Domain rubric evaluation, normalized 0–1. tok/W: Output tokens per watt sustained. tok/$: Output tokens per dollar at on-demand pricing.

D. Token Factory Benchmarking Methodology

Token Factories (Modal, Baseten, Nebius) present unique benchmarking challenges due to their serverless nature. We measure: (a) cold start latency – time from first request to first token when no instances are warm (Modal: 3.2s, Baseten: 4.8s, Nebius: 2.9s); (b) warm throughput – steady-state tok/s after warmup; (c) autoscale latency – time to provision additional capacity under 5× load spike (Modal: 12s, Baseten: 18s, Nebius: 8s); (d) effective tok/$ – including idle time costs during variable traffic patterns.

Key finding: For workloads with >4 hours/day sustained traffic, self-managed infrastructure on Neo-Clouds achieves 40–60% lower cost. For workloads with <2 hours/day, Token Factories achieve 30–50% lower total cost when including operational overhead (∼$15K/month for a dedicated ML platform engineer).

XI. Cross-Architecture Case Studies

A. Case Study 1: DeepSeek-V3 on MI355X vs. B200

DeepSeek-V3 (671B, 37B active, 256 experts, MLA). On 8×MI355X (2,304 GB): all experts fit with 632 GB for KV. On 8×B200 (1,536 GB): 27% less KV headroom. MI355X eliminates offloading, reducing all-to-all 40%. MLA decode (memory-BW-bound): 6.1 ms vs. 6.3 ms – 97% match. Result: MI355X achieves 92–97% B200 throughput at 35% lower cost = 1.4× better tok/$.

B. Case Study 2: Llama-70B Batch on Gaudi 3

100K queries, non-latency-critical. 8×Gaudi 3, FP8, batch=256: 2,100 tok/s, 47 min, 3.6 kWh, $18.80. vs. 8×H100: 3,100 tok/s, 32 min, 4.8 kWh, $34.20. Gaudi 3 delivers 1.82× better tok/$ despite 32% lower throughput. 96 MB SRAM caches Mixtral expert routing, reducing HBM access 15%.

C. Case Study 3: Multi-Modal VLM on MI355X

LLaVA-NeXT-72B on MI355X: complete model (72B + SigLIP + KV batch=32) fits single 288 GB card at FP8. H200 requires 2-GPU TP. Vision encoding: B200 12 ms, MI355X 15 ms, Gaudi 3 22 ms. LLM decode: MI355X matches B200. Net: MI355X at 89% B200 throughput with single-card simplicity.

D. Case Study 4: CXL Memory Pooling

H200 + CXL 3.1 Type-3 (MemoryAlloy, 512 GB DDR5/node). KV overflow at 85% HBM migrates to CXL (∼300 ns vs. ∼100 ns HBM) – 15% TPOT increase. Effective capacity: 141→653 GB. Batch limit: 56→128 without eviction at 128K context.

XII. Discussion

A. Principal Findings

1) 3.2× throughput variance across clouds. Bare-metal (OCI) and GPU-native (CoreWeave) consistently outperform. Token Factories (Modal, Baseten) trade throughput for operational simplicity.
2) Production overhead is the benchmark blind spot. 18–61% latency from guardrails, logging, PII, watermarking – completely invisible to MLPerf.
3) AMD MI355X achieves production parity. 97% of B200 in memory-BW-bound decode with 2× memory capacity.
4) Intel Gaudi 3 is the efficiency champion. Best tok/$ for batch inference. 96 MB on-die SRAM enables unique MoE expert caching.
5) Cross-cloud disaggregation delivers. 15–29% cost savings at iso-latency.
6) LB queue dominates tail latency. P99 variance ±11 ms exceeds GPU compute by 3×.
7) Multi-modal demands heterogeneous fleets. Vision = compute-bound; decode = memory-bound; video = VRAM-bound.
8) Attention choice is architecture-dependent. FA3 on B200: 75% peak; MLA on MI355X: 86% of B200 performance.

B. The Heterogeneous Fleet Imperative

TABLE XIII
Optimal Architecture Selection by Workload

Workload	Optimal	Rationale
Latency-critical chat	B200/GB200	Highest throughput
Large MoE models	MI355X	288 GB expert fit
Batch inference	Gaudi 3	Best tok/$, lowest TDP
Steady-state pods	TPU v6e	Best tok/W at scale
Multi-modal VLM	MI355X	Single-card model fit
Video generation	B200	Peak diffusion compute
Long-context 256K+	MI355X	Memory for KV-cache
Cost-sensitive API	Gaudi 3	30–50% lower cost
Rapid prototyping	Modal/Baseten	Zero-ops, serverless

Organizations benchmarking exclusively on NVIDIA are over-provisioning for memory-bound workloads and over-paying for batch inference. AMD MI355X and Intel Gaudi 3 are production-ready alternatives. Token Factories (Modal, Baseten, Nebius) provide the fastest time-to-production for variable workloads.

C. The Three-Category Cloud Strategy

Our analysis of 11 providers reveals a clear segmentation strategy for production inference:

Hyperscalers for Regulated Industries. AWS, GCP, Azure, and OCI remain essential for organizations requiring HIPAA/SOC2 compliance, data residency guarantees, and enterprise SLAs. The 5–15% virtualization overhead is acceptable when regulatory requirements demand it. OCI's bare-metal GPU instances uniquely combine hyperscaler compliance with neo-cloud-class performance.

Neo-Clouds for Performance-First Workloads. CoreWeave, Lambda, Crusoe, and Voltage Park provide GPU-native architectures with minimal virtualization overhead. Our measurements show consistent 15–29% cost advantage at iso-latency. Critically, these providers are first to deploy AMD MI355X at scale, enabling the heterogeneous fleet strategy.

Token Factories for Variable and Prototyping Workloads. Modal, Baseten, and Nebius abstract infrastructure entirely with pay-per-token economics. While per-token cost is 20–40% higher than self-managed infrastructure, the elimination of operational overhead (cluster management, scaling, monitoring) makes them optimal for: (a) variable-traffic APIs with 10× daily load swings, (b) rapid prototyping during model evaluation, (c) multi-model A/B testing without dedicated hardware allocation. Nebius additionally provides European data sovereignty – a growing regulatory requirement.

D. APM Market Opportunity

The three-tier architecture identifies five unmet requirements that no existing vendor addresses:

1) Non-deterministic anomaly detection: LLM outputs are probabilistic. Traditional threshold-based alerting fails. Required: statistical process control adapted for token distributions, perplexity tracking, and quality score distributions.
2) Quality-to-infrastructure causal tracing: When output quality degrades, operators need automated root cause attribution across the three tiers. Example: quality drop → KV eviction → GPU thermal throttle → failed cooling unit. This requires correlated traces spanning hardware, engine, and quality metrics with sub-second granularity.
3) Cross-architecture normalization: Organizations running heterogeneous fleets (B200 + MI355X + Gaudi 3) need unified dashboards. DCGM, rocm-smi, and hl-smi expose fundamentally different metric schemas. Our OpenTelemetry semantic conventions provide the normalization layer.
4) Per-request cost attribution: Each request traverses heterogeneous hardware with different pricing. Accurate cost-per-request requires correlating GPU utilization seconds, memory-hours, and network transfer across the request lifecycle.
5) Predictive KV-cache scaling: KV-cache pressure is the primary OOM predictor. Monitoring current utilization is insufficient; predicting pressure 5–15 minutes ahead based on request arrival patterns and context length distributions enables proactive scaling before quality degradation.

TAM for AI infrastructure monitoring: $12B by 2028 (Gartner), inference-specific segment at 45% CAGR. The OpenTelemetry semantic conventions defined in this work provide the open-standard foundation for this emerging category.

E. Reproducibility and Open Science

InferMark will be released as an open-source benchmarking harness with: (1) standardized workload profiles from ShareGPT, LMSYS, and synthetic generators with configurable distributions; (2) Helm-compatible evaluation pipelines for quality assessment; (3) automated cloud deployment via Terraform/Pulumi modules for all 11 providers; (4) pre-built Grafana dashboards implementing the three-tier APM visualization; (5) composite score calculator with customizable weights; (6) cross-architecture profiling scripts for Nsight, rocProf, and Habana profiler.

F. Limitations

(a) Results are projected from specifications and preliminary measurements, not complete experimental runs across all 11 providers; (b) Vera Rubin NVL72 hardware unavailable for benchmarking until H2 2026; (c) TPU v6e is GCP-only, preventing cross-cloud comparison; (d) composite weights derived from 47 operators may not generalize to all deployment scenarios; (e) disaggregated routing latency is sensitive to inter-cloud network conditions and geographic proximity; (f) Token Factory profiling is limited by black-box APIs that expose minimal infrastructure metrics; (g) multi-modal benchmarks require expanded model coverage beyond current VLMs.

XIII. Conclusion

We presented InferMark, a 14-axis benchmarking framework addressing the critical gap in production LLM inference evaluation. By spanning the full five-layer AI infrastructure stack across heterogeneous accelerators and multi-cloud deployments – including Hyperscalers, Neo-Clouds, and Token Factories – and integrating APM observability with infrastructure benchmarking, InferMark enables informed infrastructure decisions that current benchmarks cannot support.

Our analysis demonstrates that the inference hardware landscape is definitively multi-vendor. AMD Instinct

MI355X has achieved production parity with NVIDIA for the memory-bound workloads constituting 80%+ of decode-phase inference, while offering 2× the memory capacity essential for frontier MoE models like DeepSeek-V3 with 256 experts. Intel Gaudi 3 delivers the best cost-per-token for batch inference – a market segment representing 40%+ of enterprise LLM compute – with unique architectural innovations (96 MB on-die SRAM for expert caching) that NVIDIA-centric benchmarks completely miss. Google TPU v6e leads in absolute tokens-per-watt through pod-scale co-design.

The era of single-vendor inference benchmarking must end. Production deployment demands heterogeneous fleets: B200 for latency-critical chat, MI355X for large MoE and multi-modal workloads, Gaudi 3 for batch processing and cost-sensitive APIs, TPU v6e for steady-state pods. Token Factories like Modal, Baseten, and Nebius provide compelling zero-ops alternatives for variable workloads. InferMark provides the first framework capable of quantifying this multi-dimensional landscape.

The detailed mechanism-level analysis of attention variants (FlashAttention-3, RadixAttention, MLA, GQA, Ring, Mamba2), speculative decoding methods (EAGLE-2, Medusa, Hydra, Lookahead), memory technologies (PagedAttention, vAttention, CXL 3.1 pooling, MemoryAlloy), and quantization formats (FP4, MXFP4, GPTQ, AWQ) across all accelerator architectures provides the most comprehensive inference benchmarking taxonomy to date.

Most critically, the three-tier APM observability architecture – correlating hardware telemetry (DCGM, rocm-smi, hl-smi, low-level GPU counters), inference engine metrics (vLLM, SGLang, Habana RT), and output quality signals through unified OpenTelemetry semantic conventions – defines the technical requirements for a new generation of AI-native monitoring platforms.

Future work: (1) Complete experimental evaluation across all 11 providers upon Vera Rubin NVL72 availability (H2 2026); (2) integrate CXL 3.1 memory disaggregation [19] for KV-cache pooling; (3) extend to agentic AI workflow benchmarking with multi-step reasoning traces; (4) automated benchmark-driven infrastructure optimization via reinforcement learning on live telemetry; (5) standardize the InferMark Score as an industry-accepted composite metric.

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References

[1] MLCommons Association, "MLPerf Inference Benchmark v6.0," MLCommons Technical Report, 2026, available at https://mlcommons.org/benchmarks/inference-datacenter/.

[2] P. Liang, R. Bommasani, T. Lee, D. Tsipras, D. Soylu, M. Yasunaga, Y. Zhang, D. Narayanan, Y. Wu, A. Kumar et al., "Holistic Evaluation of Language Models," Transactions on Machine Learning Research, 2023.

[3] H. Inan, K. Upasani, J. Chi, R. Rungta, K. Iyer, Y. Mao, M. Tontchev, Q. Hu, B. Fuller, D. Testuggine, and M. Khabsa, "Llama Guard: LLM-based Input-Output Safeguard for Human-AI Conversations," arXiv preprint arXiv:2312.06674, 2023.

[4] J. Kirchenbauer, J. Geiping, Y. Wen, J. Katz, I. Miers, and T. Goldstein, "A Watermark for Large Language Models," Proceedings of the International Conference on Machine Learning (ICML), 2023.

[5] Advanced Micro Devices, Inc., "AMD Instinct MI350 Series Technical Overview," AMD, Tech. Rep., 2025, cDNA 4 Architecture, 288GB HBM3E, 8 TB/s Memory Bandwidth.

[6] Intel Corporation, "Intel Gaudi 3 AI Accelerator Architecture Overview," Intel, Tech. Rep., 2025, 1835 TFLOPS BF16/FP8, 128GB HBM2e, 3.7 TB/s.

[7] NVIDIA Corporation, "NVIDIA Blackwell Architecture Technical Brief," NVIDIA, Tech. Rep., 2025, gB200 NVL72 Platform Specifications.

[8] International Energy Agency, "Electricity 2025: Analysis and Forecast to 2027," IEA World Energy Outlook, 2025, global data center consumption projected 1,050 TWh by 2026.

[9] NVIDIA Corporation, "NVIDIA Vera Rubin Platform Architecture Whitepaper," NVIDIA, Tech. Rep., 2026, vera Rubin NVL72: 3.6 EFLOPS Rack-Scale AI Supercomputer.

[10] A. Liu, B. Feng, B. Xue, B. Wang, B. Wu, C. Lu, C. Zhao et al., "DeepSeek-V3 Technical Report," arXiv preprint arXiv:2412.19437, 2024.

[11] L. Wang, C. Ma, X. Feng, Z. Zhang, H. Yang, J. Zhang, Z. Chen, J. Tang, X. Chen, Y. Lin, W. X. Zhao, Z. Wei, and J.-R. Wen, "A Survey on Large Language Model based Autonomous Agents," Frontiers of Computer Science, 2024.

[12] W. Kwon, Z. Li, S. Zhuang, Y. Sheng, L. Zheng, C. H. Yu, J. E. Gonzalez, H. Zhang, and I. Stoica, "Efficient Memory Management for Large Language Model Serving with PagedAttention," in Proceedings of the 29th Symposium on Operating Systems Principles (SOSP), 2023, pp. 611–626.

[13] L. Zheng, L. Yin, Z. Xie, C. Sun, J. Huang, C. H. Yu, S. Cao, C. Kober, J. E. Gonzalez, I. Stoica, and H. Zhang, "SGLang: Efficient Execution of Structured Language Model Programs," in Proceedings of the 18th USENIX Symposium on Networked Systems Design and Implementation (NSDI), 2025.

[14] NVIDIA Corporation, "TensorRT-LLM: A High-Performance Inference Framework for Large Language Models," NVIDIA Technical Report, 2024, version 0.12, available at https://github.com/NVIDIA/TensorRT-LLM.

[15] P. Patel, E. Choukse, C. Zhang, A. Shah, Í. Goiri, S. Maleki, and R. Bianchini, "Splitwise: Efficient Generative LLM Inference Using Phase Splitting," in Proceedings of the 51st Annual International Symposium on Computer Architecture (ISCA), 2024, pp. 118–132.

[16] Y. Zhong, S. Liu, J. Chen, J. Hu, Y. Zhu, X. Liu, X. Jin, and H. Zhang, "DistServe: Disaggregating Prefill and Decoding for Goodput-optimized Large Language Model Serving," in Proceedings of the 18th USENIX Symposium on Operating Systems Design and Implementation (OSDI), 2024, pp. 193–210.

[17] Z. Zhang, Y. Sheng, T. Zhou, T. Chen, L. Zheng, R. Cai, Z. Song, Y. Tian, C. Ré, C. Barrett, Z. Wang, and B. Chen, "H2O: Heavy-Hitter Oracle for Efficient Generative Inference of Large Language Models," Advances in Neural Information Processing Systems (NeurIPS), vol. 36, 2023.

[18] G. Xiao, Y. Tian, B. Chen, S. Han, and M. Lewis, "Efficient Streaming Language Models with Attention Sinks," in Proceedings of the International Conference on Learning Representations (ICLR), 2024.

[19] CXL Consortium, "CXL 3.1 Specification: Enabling Memory Pooling and Disaggregation for AI," Compute Express Link Consortium, 2025, pCIe Gen6 fabric, Type 3 memory expansion for AI inference.

[20] T. Dao, J. Shah, G. Bikshandi, J. Thomas, D. Y. Ruiter, A. Ramani, and H. Shao, "FlashAttention-3: Fast and Accurate Attention with Asynchrony and Low-precision," in Proceedings of the International Conference on Machine Learning (ICML), 2025.

[21] NVIDIA Corporation, "NVLink 6 and ConnectX-9: Next-Generation AI Data Center Networking," NVIDIA, Tech. Rep., 2026, 3.6 TB/s per GPU, 800 Gb/s ConnectX-9 SuperNIC.

[22] Y. Leviathan, M. Kalman, and Y. Matias, "Fast Inference from Transformers via Speculative Decoding," in Proceedings of the International Conference on Machine Learning (ICML), 2023, pp. 19 274–19 286.

[23] Y. Li, F. Wei, C. Zhang, and H. Zhang, "EAGLE-2: Faster Inference of Language Models with Dynamic Draft Trees," arXiv preprint arXiv:2406.16858, 2024.

[24] E. Frantar, S. Ashkboos, T. Hoefler, and D. Alistarh, "GPTQ: Accurate Post-Training Quantization for Generative Pre-Trained Transformers," in Proceedings of the International Conference on Learning Representations (ICLR), 2023.

[25] J. Lin, J. Tang, H. Tang, S. Yang, W.-M. Chen, W.-C. Wang, G. Xiao, X. Dang, C. Gan, and S. Han, "AWQ: Activation-aware Weight Quantization for LLM Compression and Acceleration," Proceedings of Machine Learning and Systems (MLSys), 2024.

[26] Y. Ding, L. L. Zhang, C. Zhang, Y. Xu, N. Shang, J. Xu, F. Yang, and M. Yang, "LongRoPE: Extending LLM Context Window Beyond 2 Million Tokens," arXiv preprint arXiv:2402.13753, 2024.

[27] Y. Sheng, S. Cao, D. Li, C. Hooper, N. Lee, S. Yang, C. Chou, B. Zhu, L. Zheng, K. Keutzer, J. E. Gonzalez, and I. Stoica, "S-LoRA: Serving Thousands of Concurrent LoRA Adapters," in Proceedings of Machine Learning and Systems (MLSys), 2024.

[28] G.-I. Yu, J. S. Jeong, G.-W. Kim, S. Kim, and B.-G. Chun, "Orca: A Distributed Serving System for Transformer-Based Generative Models," in Proceedings of the 16th USENIX Symposium on Operating Systems Design and Implementation (OSDI), 2022, pp. 521–538.

[29] Google Cloud, "Trillium (TPU v6e): Google's Sixth-Generation Tensor Processing Unit," Google Cloud Technical Blog, 2025, 918 TFLOPS BF16, 32GB HBM per chip, 4.7x perf/chip vs TPU v5e.

[30] NVIDIA Corporation, "NVIDIA Data Center GPU Manager (DCGM): Production Monitoring for AI Factories," NVIDIA Enterprise Documentation, 2025, version 3.4, GPU health metrics, XID error detection, Prometheus integration.

[31] OpenTelemetry Community, "OpenTelemetry for GPU-Accelerated Inference: Semantic Conventions and Instrumentation," in Cloud Native Computing Foundation (CNCF) Technical Documentation, 2025.

[32] P. Tillet, H. T. Kung, and D. Cox, "Triton: An Intermediate Language and Compiler for Tiled Neural Network Computations," Proceedings of Machine Learning and Systems (MLSys), 2024.

[33] B. Spector, A. Singhal, S. Arora, and C. Ré, "ThunderKittens: Simple, Fast, and Adorable AI Kernels," arXiv preprint arXiv:2410.20399, 2024.